AIG CSAM and Children's Safety

AIG CSAM?



CSAM refers to Child Sexual Abuse Material, with AIG CSAM being Artificial Intelligence Generated Child Sexual Abuse Material. AIG CSAM has been a rising problem in the AI world, with 1 in 29 children worldwide experiencing sexual extortion(which is when a person is blackmailed with sexual material).

AIG CSAM can be categorized as either fully synthetic or partially synthetic. Fully synthetic meaning that the image was generated entirely by AI; Partially synthetic meaning that an existing image was used to create illegal material(such a the nudification or face swapping of photos of existing people).





There are many issues with AIG CSAM in terms of prevention and mitigation. AIG CSAM can be produced en masse and without the victim's knowledge while maintaining its photo realism. The overwhelming number of AIG CSAM prevents effective tracking of illegal material. In addition, AI impedes victim identification, because synthetic people and real children are difficult to differentiate.

AIG CSAM has also accelerated re-victimization and victimization as new photos and videos are easily generated by AI as long as photos of the child is available to the criminal generating the CSAM. These generative AI tools have become much more common, so much so that nudifying peers has become a new form of bullying among the younger generation. Instagram recently released and withdrew their most recent AI feature that allowed users to mention public Instagram accounts to Meta AI to generate images of the photos in those accounts. It was taken down promptly due to backlash.





AIG CSAM also drive the black market. In the past, illicit material wasn't efficient to generate to earn profits. However, due to AIG CSAM being easy to generate en masse, profit driven ecosystems like the black market become much more prolific.

Mitigation Strategies for AIG CSAM

Current Mitigation Strategies



Current mitigation strategies include:

- Hashing and Matching

A process that translates files into unique codes called hashes, which are then compared to existing databases of known hashes to detect illegal material.

- Notice and Take down

Is the process where victims or users can report illicit content for it to be taken down legally.

- ML/AI Classifiers

Classifiers is when AI systems are trained to categorize content into different labels such as spam and not spam for emails.

However, none of the currents strategies being used can keep up with AIG CSAM in the speed and precision that it requires. They do not work on the mass scale that they need to. In addition, all of the current existing strategies are reactive, not preventative, meaning that the illicit content has already been made before being detected. Images released on the internet often have copies floating around in the online space.





Potential ways that mitigation strategies can be improved are:

- Train AI so that they don't collect and train on data from known CSAM sites

There are always known CSAM sites that are difficult to take down, this means that AI developers can make sure that the AI is not trained from data that originates from these websites.

- Use the following assessment strategies:
  - CSEA Assessment

CSEA is an assessment that looks for predatory behavior such as online grooming, live stream abuse and peer-to-peer solicitation etc. This is a preventative strategy that stops CSAM before it happens.

  - Assess 3rd Party Models for Harmful Capabilities before Implementing

If 3rd Party AI tools are being used for certain capabilities, assessments on whether the AI has CSAM capabilities should be tested.



The limitation of these preventative CSAM testing is that generating illicit material is illegal, even if it is for testing. Therefore developers are unable to know 100% sure if their AI model does not generate CSAM.



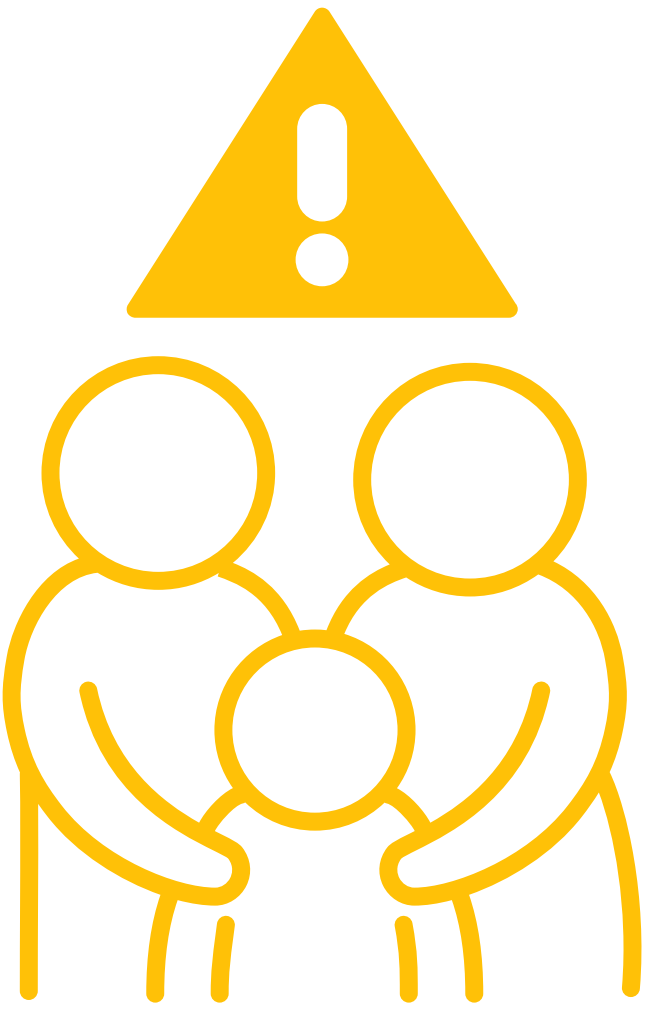
Sharenthood and Minor Protection on the Internet

Parents on the Internet



With the rise of the internet, more and more parents are sharing photos of their children on the internet. Whether it be to catch up with relatives and friends or for an actual influencer business, posting photos and videos of young children has become normalized. This, however, means that these children are much more vulnerable to sexual extortion, specifically AIG CSAM.

The primary role of parents is that they are legal guardians responsible for keeping their children safe. They are the primary guardians against abuse such as sexual extortion. However, parents are unable to do so due to them not knowing the consequences of uploading photos and videos of their children on the internet.
It has been found to be even worse for the older generations AKA the grandparents. The older generation has become much more active on social media due many retiring and having increasing time to invest in the internet. Grandparents are also increasingly likely to upload content of their grandchildren on the internet, with even less understanding of how it can impact the child under their protection.



Children Internet Exposure



The age in which children are introduced to digital devices and most importantly, the internet is becoming lower and lower. Some children are introduced to social media platforms like youtube starting from the age of 2. From the ages 3~18, children are still malleable, meaning that their identities are still being formed and they are developing complex emotions and thoughts. Specifically, the ages 12~18 are critical for forming independent thought and identities, while learning to understand social cues.

The early interactions with AI to children may have a critical impact on their developmental process. Due to the lack of social understanding children have at a young age, interacting with sycophantic AI results in children forgetting how to be polite, and missing important social cues for empathy. This also brings up the question of trust and relationship building, and concerns that children may form deeper emotional attachments to AI than to real people.

